

Dual Graph Network Hashing for Cross-Modal Retrieval

Shuang Zhang[✉], Yue Wu, Lei Shi[✉], Feifei Kou[✉], Huilong Jin[✉], Pengfei Zhang[✉], *Associate Member, IEEE*,
Weiping Ding[✉], Mingying Xu, and Muhammet Deveci[✉]

Abstract—Hashing algorithms represent data by generating compact binary hash codes, enabling efficient cross-modal similarity search and significantly improving the storage efficiency and retrieval performance of image-text data. However, because traditional hashing methods typically separate image-text feature extraction from hash learning, the feature extraction module struggles to adaptively update based on training feedback, further limiting the performance of cross-modal retrieval in real-world scenarios. To address this issue, deep learning has been introduced to cross-modal hashing, enabling end-to-end joint optimization and tightly integrating feature extraction and hash learning, significantly improving retrieval performance. However, because existing deep learning methods often use a fixed weight distribution when processing samples, they ignore the modal differences between text and visual features when fusing them. This leads to an inadequate fused representation and difficulty achieving optimal modality

alignment during hash code generation. To address this issue, we propose a Dual Graph Network Hashing (DGNH) algorithm that dynamically adjusts the weight distribution between visual and text features through an adaptive attention mechanism, ensuring better modality fusion during hash code generation. Specifically, we design a novel framework that combines a graph convolutional neural network (GCN) with a graph attention network (GAT) to construct a label classifier for generating labels and enhancing cross-modal feature representation. This approach improves feature discrimination by capturing the hierarchical relationships and co-occurrence patterns of labels through a carefully constructed label association graph. Furthermore, we introduced a pre-trained model combining CLIP and the Transformer to further enhance the overall feature representation. During the optimization phase, we employed a contrastive triplet loss function coupled with novel regularization constraints for quantization and optimization, thus effectively reducing information loss during discretization and ensuring the generated hash codes are more compact and efficient. Experimental results on three public datasets, MS-COCO, NUS-WIDE, and MIRFlickr-25 K, demonstrate that the proposed method outperforms existing methods in both retrieval accuracy and efficiency, thereby validating its effectiveness and superiority.

Index Terms—Cross-modal hashing retrieval, adaptive attention mechanisms, graph convolutional network, graph attention network.

I. INTRODUCTION

WITH the rapid advancement of information technology and the widespread proliferation of multimedia content, single-modal data retrieval is increasingly unable to satisfy users' diverse information needs. As a result, the ability to efficiently acquire and process cross-modal data—such as text, audio, video, and images—has emerged as a significant focus within the research community. Multi-view learning provides core theoretical and methodological support for solving heterogeneous multimedia data analysis problems, which has formed a systematic and mature research system after decades of development [1]. Further explorations on relational knowledge modeling among different views effectively promote the semantic correlation mining of cross-modal heterogeneous data [2]. Conventional single-modal retrieval approaches struggle to bridge the semantic gap between different modalities, thereby limiting their effectiveness in addressing the complex demands of modern applications [3], [4], [5]. Consequently, cross-modal retrieval has become a pivotal area of research in information retrieval. The fundamental objective of cross-modal retrieval is to establish semantic alignment among heterogeneous data types, allowing users to retrieve relevant content across modalities using queries

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Shuang Zhang, Yue Wu, and Huilong Jin are with the College of Engineering, Hebei Normal University, Shijiazhuang 050024, China (e-mail: zshuang@hebtu.edu.cn; wuyue@stu.hebtu.edu.cn; JHL981@hebtu.edu.cn).

Lei Shi is with the State Key Laboratory of Media Convergence and Communication, Communication University of China, Beijing 100024, China (e-mail: leiky_shi@cuc.edu.cn).

Feifei Kou is with the School of Computer Science(National Pilot Software Engineering School), Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: koufeifei000@bupt.edu.cn).

Pengfei Zhang is with the School of Computer Science and Engineering, Anhui University of Science and Technology, Huainan 232001, China (e-mail: zpf.bupt@bupt.cn).

Weiping Ding is with the School of Artificial Intelligence and Computer Science, Nantong University, Nantong 226019, China (e-mail: dwp9988@163.com).

Mingying Xu is with the School of Artificial Intelligence and Computer Science, North China University of Technology, Beijing 100144, China (e-mail: xumingying@ncut.edu.cn).

Muhammet Deveci is with the Department of Industrial Engineering, Turkish Naval Academy, National Defence University, 34942 Tuzla, Türkiye, also with the Sogang University, Seoul 04107, Republic of Korea, also with the Department of Information Technologies, Western Caspian University, Baku 1001, Azerbaijan, and also with the Department of Computer Science and Engineering, College of Informatics, Korea University, Seoul 02841, Republic of Korea (e-mail: muhammetdeveci@gmail.com).

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from any single modality. In pursuit of this goal, numerous approaches have been proposed, with the integration of deep learning and hash learning standing out as the most promising and efficient solutions. Deep learning methods are capable of automatically learning high-level semantic representations, thereby overcoming the limitations of traditional approaches that rely on handcrafted features. Meanwhile, hash learning narrows the modality gap by encoding samples into compact binary hash codes, which also substantially increases the efficiency of storage and retrieval operations [6], [7], [8], [9], [10].

Cross-modal hashing techniques are generally divided into two main categories based on whether semantic label information is utilized: unsupervised cross-modal hashing [11], [12], [13], [14] and supervised cross-modal hashing [15], [16], [17], [18], [19]. Unsupervised hashing methods generate hash codes by mining intrinsic associations within the data itself, without relying on semantic labels. In contrast, supervised approaches employ label information to enforce semantic consistency across different modalities, thereby more precisely modeling the semantic relationships among them. In practice, unsupervised methods often struggle to capture complex semantic structures and intricate cross-modal interactions. Supervised methods, owing to their higher semantic accuracy, have become a prominent area of research and are the primary focus of this paper.

Among the cross-modal retrieval methods based on deep learning, Graph Neural Networks (GNNs) [20], [21], [22] have been widely regarded as the most representative models in recent years due to the introduction of graph structures. Their advantage lies in the ability to construct semantic graphs and model the semantic relationships between nodes through edge representations, thereby achieving cross-modal semantic alignment. Additionally, GNNs can be combined with deep learning structures, such as convolutional neural networks and attention mechanisms, to further enhance cross-modal retrieval performance. Among them, the most representative are Graph Convolutional Networks (GCNs) [23], [24], [25] and Graph Attention Networks (GATs) [26], [27]. GCN extends convolution operations to graph structures and obtains robust node representations through neighborhood aggregation, while GAT incorporates attention mechanisms, enabling the model to adaptively assign neighbor weights and thereby highlight the most relevant node information.

However, while cross-modal methods based on deep learning and hashing learning, particularly those combined with GNNs, have made remarkable progress, several challenges remain. On the one hand, some existing methods often overlook the potential inherent structural features in different modal data when learning a unified cross-modal representation, thereby failing to fully explore and utilize the semantic information of multimodal data. On the other hand, many supervised methods rely on label information to generate correlation matrices, a process that is not only computationally complex but also difficult to scale for large datasets with labels, thereby limiting their practical application. Consequently, more efficient semantic correlation modeling mechanisms and more computationally efficient label utilization methods have become key challenges in improving the performance and practicality of cross-modal retrieval. To address these

challenges, we propose a new supervised cross-modal hashing method that fully explores and utilizes the semantic correlation of cross-modal data while reducing computational reliance on label construction. Our research not only provides an efficient and feasible new idea for cross-modal retrieval in theory, but also offers a potential solution for the practical application of large-scale multimodal data retrieval.

This paper introduces DGNH, a graph network-based supervised cross-modal hashing method designed to address the aforementioned challenges. Unlike conventional multimodal representation learning methods, DGNH jointly integrates semantic alignment, graph-based label reasoning, and quantization-aware hash learning within a unified cross-modal hashing framework. Our approach employs a three-part framework for cross-modal retrieval tasks, consisting of feature extraction, label classification, and hash code generation. First, in the feature extraction stage, we combine a Transformer model with the Contrastive Language-Image Pre-training (CLIP) model to simultaneously obtain global feature representations for both image and text modalities. This module captures richer and higher-level semantic information, providing a solid feature foundation for subsequent retrieval. Subsequently, in the label classification module, we introduce a Graph Convolutional Network (GCN) and a Graph Attention Network (GAT) to construct the label matrix. Specifically, the GCN effectively captures global semantic relationships between labels by performing neighborhood aggregation on the label graph, while the GAT leverages an attention mechanism to assign adaptive weights to different neighboring labels, thereby highlighting the most relevant label relationships for the task at hand. Through the synergistic effect of these two graph neural networks, the label classifier not only enhances semantic modeling capabilities but also significantly reduces the computational complexity associated with labeling, laying the foundation for efficient retrieval. Finally, in the hash code module, we constructed two functions to ensure the effectiveness and stability of the generated hash codes. On the one hand, we introduced a comparative triplet loss function to shorten the hash distance between semantically similar samples and increase the hash distance between dissimilar samples, thereby improving the robustness and discriminability of the hash code. On the other hand, we combined it with a cosine similarity function, making the hash code more compact while preserving cross-modal semantic information more effectively. The following are our primary contributions:

- We develop a strong feature extraction approach for multimodal data using Transformer and Contrastive Language-Image Pre-training (CLIP) models. For image processing, we utilize both CLIP's visual encoder and a Transformer backbone to capture rich visual representations.
- For text data, we employ a Transformer architecture augmented with a contextual attention mechanism. This design strengthens inter-modal feature connections, enabling our end-to-end framework to effectively align and enhance global representations while maintaining semantic consistency among images and text.
- To find hidden information in labels and provide predictions, we present a label classifier that is built on graph

convolutional networks and graph attention networks. The feature extraction module extracts features and combines them with the expected labels for enhanced feature representation.

- We evaluated our approach on three standard benchmarks (MS-COCO, NUS-WIDE, and MIRFlickr-25 k), where it surpassed multiple leading cross-modal hashing methods and achieved over 93% accuracy on the Top-1000 curve for all datasets.

The rest of this paper is organized as follows: Section II reviews the relevant prior research, summarizing key advances and identifying unresolved challenges in the field. Section III details the proposed DGNH approach, introducing its novel architecture and explaining how each component addresses existing limitations. Section IV presents experimental results, demonstrating the effectiveness of the approach in comparison to cutting-edge methods on benchmark datasets. Section V highlights contributions and presents ideas for future study areas.

II. RELATED WORK

A. Supervised Cross-Modal Hashing

To enhance retrieval effectiveness and precision, supervised cross-modal hashing [15], [16], [28], [29], [30] maps data from various modalities into a common hash space through supervised learning, learns the semantic associations between data from different modalities using label information or supervisory signals, and utilizes hash functions to convert high-dimensional information into low-dimensional binary hash codes. Traditional supervised cross-modal hashing techniques mostly rely on superficial models or linear projection, which learn semantic correlations between various modal data and map them into a common hash space using label information or supervision signals. For instance, Semantic Correlation Maximization (SCM) [31] embeds semantic labels directly into the hashing method for effective large-scale data modeling, utilizing all available supervised data to achieve linear-time training. Semantic-preserving Hashing (SePH) [32] approximates the supervision information using the hash code obtained from the Hamming space by minimizing the KL divergence, while maintaining semantic information and converting it into a probability distribution.

However, traditional supervised methods are typically based on linear projection or shallow models, which struggle to capture complex, nonlinear relationships. At the same time, they require a substantial amount of similarity matrix or label information, which limits their application in unsupervised or weakly supervised scenarios. Deep cross-modal hashing [33], [34], [35] approaches have progressively supplanted conventional approaches as deep learning has advanced. By having neural networks automatically learn both feature representations and hash functions, these methods more effectively model nonlinear relationships and semantic content in high-dimensional data. Deep Cross-Modal Hashing (DCMH) [33] is a deep neural network-based end-to-end learning framework that combines feature and hash code learning into a single framework. It can learn discrete hash codes directly without the need for relaxation. The first to employ adversarial learning for cross-modal hashing in a self-supervised manner is Self-Supervised

Adversarial Hashing (SSAH) [34], which maximizes the semantic relevance and consistency between various modalities using two adversarial networks and utilizes a self-supervised semantic network to discover high-level semantic information. To overcome the modality gap, Multi-Relational Deep Hashing (MRDH) [35] models similar relationships across various modal data and creates a new similarity metric that enables the model to produce more discriminative hash codes. Recent studies have explored more refined semantic modeling strategies to improve cross-modal hashing performance. Deep Hierarchy-aware Proxy Hashing (DHaPH) [36] constructs a data-driven semantic hierarchy via learnable proxy representations, enabling the model to capture fine-grained semantic relationships and better distinguish intra-class and inter-class variations. Meanwhile, Deep Semantic-consistent Penalizing Hashing (DScPH) [37] introduces a consistent penalizing mechanism to learn unified decision boundaries across classes, addressing the limitations of traditional margin-based constraints and enhancing semantic consistency during hash learning.

B. Unsupervised Cross-Modal Hashing

The intrinsic structure or similarity of the material itself—such as similarities within and between modalities—is utilized by unsupervised cross-modal hashing techniques to produce hash codes through unsupervised learning. There are two types of unsupervised hashing techniques: deep learning techniques and conventional techniques. In traditional techniques, using a hash function, Cross-View Hashing (CVH) [38] performs cross-view similarity retrieval by mapping comparable items to similar codes across views. Latent Semantic Sparse Hashing (LSSH) [39] employs matrix factorization to extract latent concepts from text and sparse coding to extract salient patterns in images. A shared abstract space is then mapped to the latent semantic features that have been learned.

Neural networks are utilized in deep learning techniques to automatically enhance feature representations and hash functions, which are more capable of capturing semantic information from high-dimensional data and nonlinear relationships. Unsupervised Contrastive Cross-Modal Hashing (UCCH) [40] employs a novel momentum optimizer to directly perform hashing operations in contrastive learning, thereby enhancing retrieval performance through both contrastive learning and global strategies. Similarity Graph-correlation Reconstruction Network (SGRN) [41] achieves finer-grained alignment between modalities by constructing and optimizing fine-grained similarity relationship graphs as well as global relationship graphs, combining similarity representations of real-valued and hashed features to bridge the modality gap. Ensemble Prototype Networks (EPNet) [42] determines class properties of deep features, produces precise and compact hash codes, and employs ensemble clustering and aggregation of local correlations to extract correlation information between cross-modal occurrences from many perspectives.

C. Graph Convolutional Network

In recent years, convolutional neural networks have significantly advanced cross-modal hash retrieval by extracting

high-quality features, though they are limited to regular grid data. To address irregular graph-structured data, Joan Bruna et al. [43] extended convolution to the graph domain using spectral graph theory. However, the high computational cost of spectral methods led to the development of more efficient spatial [44], [45] approaches, which update node representations by aggregating neighborhood features and better capture graph structures.

Graph Convolutional Network Discrete Hashing (GCDH) [23] introduces a GCN-based classifier to extract label information and bridge modality gaps, learning discrete binary codes while preserving semantic relationships in the Hamming space. Modality-specific and Cross-modal Graph Convolutional Networks (MCGCN) [46] constructs cross-modal graphs for semi-supervised hashing, leveraging intra- and inter-modal relationships as well as labeled and unlabeled data to learn discriminative hash codes. Additionally, the Novel Dialog and Event Relation-aware Graph Convolutional Neural Network (DERGCN) [47] models complex relational dependencies through a multi-relation graph and enhances performance using self-supervised learning, transformer-based fusion, and contrastive optimization.

D. Graph Attention Network

Graph Attention Networks (GATs) are an important branch of graph neural networks. They incorporate an attention mechanism when aggregating neighbor features, adaptively assigning weights based on node characteristics rather than fixed graph structure, thus overcoming the limitation of fixed aggregation weights in GCNs. Unlike GCNs, which are based on spectral methods, GATs are non-spectral methods, possessing stronger inductive learning capabilities and more efficient processing of complex graph data.

Multi-Label Semantic Supervised Graph Attention Hashing (MSGAH) [27] enriches semantic information through an end-to-end label encoder, builds graph features through the adjacency relationship of nodes, assigns different weights to adjacent edges, and uses multi-label annotations to connect the semantic relevance between modalities. Domain-adaptive Graph Attention-supervised Network (DGASN) [48] was first proposed and studied the problem of cross-network homogeneous and heterogeneous edge classification. It adopted multi-head GAT as the graph neural network encoder, jointly trained node embeddings and edge embeddings through node classification and edge classification losses to produce informative embeddings to distinguish homogeneous and heterogeneous edges, applied direct supervision to graph attention learning to reduce the negative impact of heterogeneous edges, and learned network-invariant edge embeddings through adversarial domain adaptation.

As previously said, even though cross-modal hashing retrieval in graph networks has received plenty of attention, the methods listed above have the following drawbacks and difficulties: 1) Existing methods often fail to establish deep semantic consistency between visual and textual features due to insufficient cross-modal interaction mechanisms. 2) Current frameworks

TABLE I
SYMBOLS AND THEIR CORRESPONDING MEANINGS

Notation	Meaning of notations
$\mathbf{O} = \{(x_i, y_i) i \in [1, N]\}$	Training dataset
$\mathbf{x}_i \in \mathbb{R}^{dI}, \mathbf{y}_i \in \mathbb{R}^{dT}$	Modality i -th data
$\mathbf{D}_I \in \mathbb{R}^{dI}, \mathbf{D}_T \in \mathbb{R}^{dT}$	Modality extracted features
$\mathbf{I}_i \in \{0, 1\}^{N_l}$	Binary vector
$\mathbf{L} = \{\mathbf{I}_i i \in [1, N]\}$	Label set
$\mathbf{G} = \{\mathbf{g}_i i \in [1, N_l]\}$	Glove vector
$\mathbf{A} \in \mathbb{R}^{N_l \times N_l}$	Adjacency matrix
$\mathbf{H}^I \in \{-1, 1\}^c, \mathbf{H}^T \in \{-1, 1\}^c$	Image and text hash codes
N	Number of samples
dI, dT	Dimension of the modality features

typically process image and text features independently without robust attention-based fusion, resulting in suboptimal global representation learning. 3) Prior works usually treat labels as independent categories, overlooking the valuable topological relationships and dependencies among them that could enhance feature discrimination. 4) Existing solutions often employ disjointed pipelines for feature extraction and label prediction, missing opportunities for synergistic optimization through end-to-end learning.

III. THE PROPOSED METHOD

A. Notation and Problem Definition

The notations employed by this paper are first explained. Suppose that we have a training dataset $\mathbf{O} = \{(x_i, y_i) | i \in [1, N]\}$, whereby data from the i -th image modality is represented by $\mathbf{x}_i \in \mathbb{R}^{dI}$, and the data from the i -th text modality is represented by $\mathbf{y}_i \in \mathbb{R}^{dT}$. We use $\mathbf{D}_I \in \mathbb{R}^{dI}$ to represent the extracted features of the image modality and $\mathbf{D}_T \in \mathbb{R}^{dT}$ to represent the extracted features of the text modality. In addition, N , dI , and dT represent the quantity for samples, the dimension within the image features, and the dimension within the text features.

The labels of each instance are capable of being shown to serve as the binary vector $\mathbf{I}_i \in \{0, 1\}^{N_l}$. N_l represents the number of classes. The label set $\mathbf{L} = \{\mathbf{I}_i | i \in [1, N]\}$ is made up from the labels for every instance. We use the glove [49] vectors $\mathbf{G} = \{\mathbf{g}_i | i \in [1, N_l]\}$ and the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N_l \times N_l}$ as the inputs to the label classifier module, where $\mathbf{g}_i \in \mathbb{R}^{dl}$ represents the dl -dimensional glove vector corresponding to the i -th label, which is pre-trained on the Wikipedia dataset.

This paper's objective is to acquire hash codes $\mathbf{H}^I \in \{-1, 1\}^c$ and $\mathbf{H}^T \in \{-1, 1\}^c$ from the hash functions designed for text and images, specifically, where c denotes the hash code's length. The $\tanh(\cdot)$ and $\text{sign}(\cdot)$ functions are employed to restrict the hash codes. To enhance understanding, we summarize the above notations in Table I.

B. Network Architecture of DGNH

Our DGNH architecture, which includes two streams for learning hash functions of the text and image modalities, is displayed in Fig. 1. We obtain features from the image data by utilizing the Transformer model and the CLIP model, and then we use the Transformer model to obtain features from the text data. To dynamically allocate attention weights to every

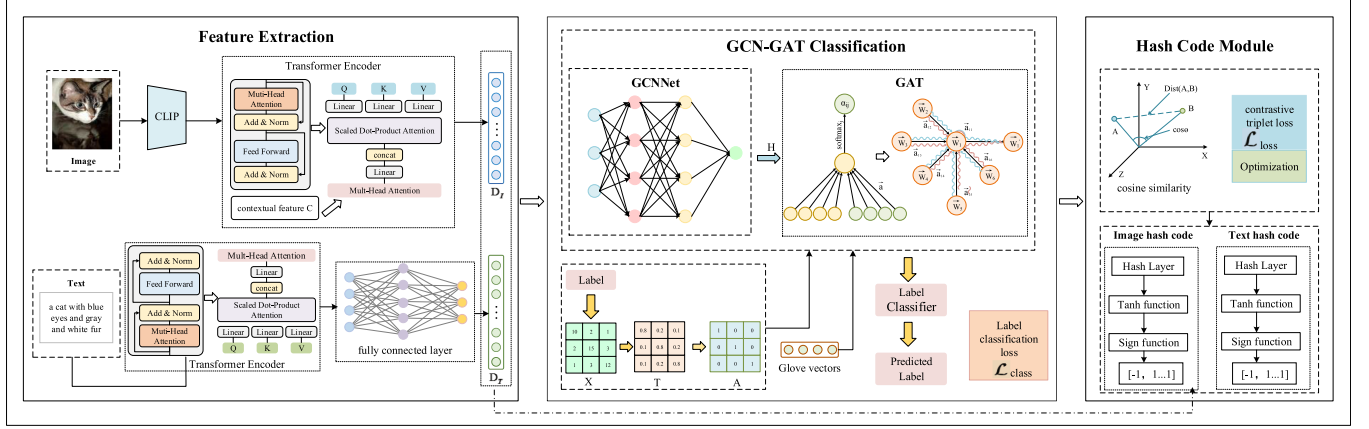


Fig. 1. Overall framework of DGNH. DGNH mainly consists of three modules: 1) Feature extraction module: utilizes CLIP and Transformer encoders to extract features from each modality; 2) classifier module: employs GCN and GAT as label classifiers to obtain rich semantic information of labels; 3) hash code module: generates hash codes through the hyperbolic tangent function, the sign function, and optimization strategies.

element within the input sequence, depending on the relationship between the current element and other components, we implemented a contextual attention mechanism throughout the image feature extraction process. This enables the model to capture contextual information more accurately, thereby improving its capability to understand and process sequential data. An initial representation of the text data is a Bag-of-Words (BoW) vector. We use a classifier module based on GCN and GAT to increase the discriminativeness of the text and image data. This module predicts labels for instances in both modalities and embeds the co-occurrence relationships among labels within the word vector representations. The predicted labels close the semantic gap among various modalities and enhance the model's semantic information. The classifier module models the semantic relationship between labels so that the model not only relies on the characteristics of a single label itself, but also utilizes the correlation between labels, thereby improving the richness and discriminability of feature representation, further enhancing feature representation and improving retrieval efficiency. Ultimately, we improve retrieval quality by obtaining the ideal hash codes through the optimization approach and the hash code loss function.

C. Feature Extraction

The feature representations based on the original text and images are obtained using two feature extraction networks in our model and used as input for the next model. We employ CLIP [50] to project both images and text into a common embedding space, enabling the quantification of their semantic relationship via vector similarity. Due to CLIP providing pre-aligned image-text embeddings that preserve semantic similarity across modalities, while the Transformer's global self-attention captures long-range dependencies and contextual relationships. Together, these properties enable the generation of semantically consistent and highly discriminative hash codes, offering a theoretical advantage over traditional unimodal architectures such as ResNet and BERT.

In our image network, the input of the image modality is denoted as $\mathbf{x}_i$, and the specific feature extraction process can be represented as:

$$f_x = \text{clip}(\mathbf{x}_i). \quad (1)$$

We utilize the Transformer encoder and the contextual attention mechanism to expand and align the dimensions, because CLIP maps images to a high-dimensional feature space. This allows the model to dynamically assign weights to each element and adaptively adjust its focus according to contextual importance, thereby enhancing semantic understanding of the input data and producing richer feature representations. Specifically, the contextual attention mechanism is implemented as a standalone module rather than being embedded within the Transformer. The Transformer first expands and refines the CLIP image embeddings, after which contextual attention computes interactions between the expanded features and contextual information to generate a context-aware representation, while preserving the original Transformer architecture.

Suppose that the contextual features are denoted as $\mathbf{C} \in \mathbb{R}^{N \times dI}$, where the number of contextual features is denoted by N . The following is a representation of the process:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right), \quad (2)$$

$$\mathbf{f}'_x = \mathbf{A}\mathbf{V}, \quad (3)$$

$$\mathbf{Q} = f_x \mathbf{W}_Q, \mathbf{K} = \mathbf{C} \mathbf{W}_K, \mathbf{V} = \mathbf{C} \mathbf{W}_V, \quad (4)$$

where $\mathbf{A}$ denotes the attention score and $\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}$ denotes the attention scoring function. The weight matrices $\mathbf{W}_Q \in \mathbb{R}^{d \times dI}$, $\mathbf{W}_K \in \mathbb{R}^{dI \times dI}$ and $\mathbf{W}_V \in \mathbb{R}^{dI \times dI}$ are used to transform the input feature and the contextual features $\mathbf{C}$ into queries, keys, and values, respectively.

$$\text{FFN}(x) = \max(0, x \mathbf{W}_1 + b_1) \mathbf{W}_2 + b_2, \quad (5)$$

$$\mathbf{D}_I = \text{FFN}(\mathbf{f}'_x) \mathbf{W}_{out} + b_{out}. \quad (6)$$

The output of the network is processed through a feedforward network. Here, $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{W}_{out}$ represent the weight matrices for the intermediate linear layer and the feedforward network's last output layer, respectively. The terms b_1 , b_2 , and b_{out} correspond to the bias terms in these layers. The resulting feature $\mathbf{D}_I$ represents the adjusted and aligned image feature dimension.

In our model architecture, we encode the text using the Transformer model to obtain a high-dimensional feature representation, denoted by f_y .

However, although the feature representation generated by the Transformer contains rich semantic information, its dimensionality may not fully align with the requirements of downstream tasks. In effort to suitably modify the dimension of the high-dimensional characteristics that were taken from the Transformer model, we add a fully connected layer to the text network. This improves the representation capability of the model and enhances its efficiency. Using the Transformer and the fully connected layer, we obtain the extracted text features $\mathbf{D}_T$.

D. Classifier Module

The classifier module, which maps node features to category labels, is often found after the graph neural network's final layer. Given that GCN and GAT can capture the co-occurrence information and semantic associations between labels through graph structures, in this section, to incorporate label co-occurrence information within word embeddings, we utilize a label classifier based on GCN and GAT, thereby predicting the labels of modalities and further combining the predicted labels with the feature representations. So the term "dual graph" refers to the use of two graph neural network architectures (GCN and GAT) operating on the same label association graph, rather than two distinct graph structures. The GCN-GAT label classifier is not redundant to CLIP, as it captures structured label dependencies (e.g., co-occurrence and higher-order relationships) that CLIP's instance-level contrastive learning does not explicitly model. By constructing a filtered label graph and applying attention weighting, it provides complementary semantic guidance while mitigating noisy connections, thereby improving semantic consistency in hash code generation rather than introducing noise. The specific operation of generating the adjacency matrix $\mathbf{A}$ from labels is shown in Fig. 2.

To create a co-occurrence matrix $\mathbf{X}$, we first analyze the frequency that label pairings occur together in the training set. The number of times label i and label j appear together is indicated by $\mathbf{X}_{ij}$. Then, we can calculate the marginal probability $\mathbf{N}_i$ of label i :

$$\mathbf{N}_i = \sum_{j=1}^N \mathbf{X}_{ij}, \quad (7)$$

where $\mathbf{N}_i$ is the whole amount for times label i appears. The conditional probability matrix $\mathbf{T}$, where $\mathbf{T}_{ij}$ denotes a likelihood for label j appearing when label i appears, could be computed

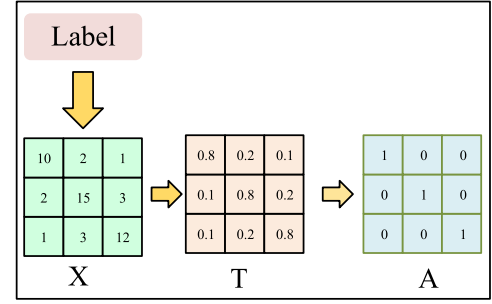


Fig. 2. Generate adjacency matrix.

through the acquired marginal probability:

$$\mathbf{T}_{ij} = \frac{\mathbf{X}_{ij}}{\mathbf{N}_i}. \quad (8)$$

Finally, we set a threshold λ to filter out edge connections that may be caused by noise, generating the adjacency matrix $\mathbf{A}$. The threshold λ is treated as a hyperparameter and selected via validation by searching a small range (e.g., 0.005–0.05) and choosing the value that yields the best mAP. By suppressing weak semantic correlations and preserving reliable label associations, the proposed graph construction strategy improves label relationship modeling and maintains robustness to noisy or incomplete labels. Moreover, retrieval performance remains relatively stable under moderate threshold variations. The process is as follows:

$$\mathbf{A}_{ij} = \begin{cases} 0 & \text{if } \mathbf{T}_{ij} < \lambda, \\ 1 & \text{if } \mathbf{T}_{ij} \geq \lambda. \end{cases} \quad (9)$$

The adjacency matrix $\mathbf{A}$ and the input feature matrix $\mathbf{X}$ are utilized by the GCN layer to accomplish feature fusion. The layer-by-layer propagating rule of GCN is as follows:

$$\mathbf{H}^{l+1} = \sigma(\tilde{\mathbf{A}} \mathbf{G}^l \mathbf{W}^l), \quad (10)$$

where $\mathbf{G}^l$ is the glove vector, $\mathbf{W}^l$ is the weight of GCN layer, the normalized form for the adjacency matrix $\mathbf{A}$ is $\tilde{\mathbf{A}}$, and nonlinear operations are represented by $\sigma(\cdot)$. The label association graph is constructed from label co-occurrence statistics derived from the training data to capture semantic dependencies among labels. By generating an adjacency matrix, the resulting graph encodes both statistical correlations and underlying semantic relationships between labels.

According to the adjacency matrix $\mathbf{A}$, the known data point i 's neighborhood set of nearby points j is $\mathcal{N}(i) = \{j | \mathbf{A}_{ij} = 1, 1 \leq j \leq n\}$. Here, $\mathbf{A}_{ij} = 1$ indicates that a link exists among node i and node j .

The GAT layer assigns weights to each label pair through the attention mechanism and calculates the attention coefficient for each label pair (i, j) as follows [26]:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{H}\mathbf{g}_i \parallel \mathbf{H}\mathbf{g}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{H}\mathbf{g}_i \parallel \mathbf{H}\mathbf{g}_k]))}, \quad (11)$$

where $\mathbf{H}$ represents a learnable weight matrix, $\mathbf{a}$ is the attention vector utilized for calculating the correlation between label pairs, $\mathbf{g}_i$ and $\mathbf{g}_j$ indicate the glove vectors corresponding to labels i and j , respectively. Through attention coefficients, the GAT layer enables the adaptive weighting of neighboring labels.

Based on the obtained attention coefficient, the feature vectors of the labels are weighted and summed to update the feature representation of the labels:

$$\mathbf{H}_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{H} \mathbf{g}_j. \quad (12)$$

At the last layer of GAT, the output of feature representation is $\mathbf{H}_i$, which is used to calculate the predicted score $\mathbf{Y}_i$ of the label:

$$\mathbf{Y}_i = \mathbf{H}_i \mathbf{W}_{class}, \quad (13)$$

where $\mathbf{W}_{class}$ is the classification weight matrix. The multi-label classification loss can be expressed as:

$$\mathcal{L}_{class} = \sum_{i=1}^N [\mathbf{Y}_i \log(\sigma(\mathbf{Y}_i)) + (1 - \mathbf{Y}_i) \log(1 - \sigma(\mathbf{Y}_i))]. \quad (14)$$

In this context, N signifies the number of labels associated with the given sample. The sum of the losses for all N labels is utilized to calculate the sample's overall loss. $\mathbf{Y}_i$ signifies the sample labels' ground-truth values. The term $\log(\sigma(\mathbf{Y}_i))$ represents the predicted scores after applying the sigmoid function, which maps the raw scores into the interval (0,1). The logarithm is employed to calculate the cross-entropy loss. By modeling the correlation between labels through graph structure, the model can more accurately predict the probability of each label appearing, thereby improving classification accuracy. Send the predicted labels to the hash layer for the next step of operation. GCN and GAT are employed not only for label classification but also to learn structured semantic label representations that guide hash code learning. A label co-occurrence graph is constructed from conditional label probabilities, with GCN propagating semantic information among correlated labels to capture higher-order dependencies, and GAT refining the process via adaptive attention weights that emphasize informative relationships and reduce noise. These representations can enhance semantic similarity preservation in the hash space, yielding more discriminative and semantically consistent hash codes.

E. Hash Code Module

In this section, we achieved the mapping of continuous features to discrete hash codes by combining the $\tanh(\cdot)$ and $\text{sign}(\cdot)$ functions and comparing the strategies of triplet loss and cosine similarity, which improves the accuracy and stability of feature representation and helps to learn semantic similarity.

1) *Hash Function*: During training, overfitting and vanishing hash code gradients may occur. Overfitting is usually caused by an overly complex model or insufficient training data, while the vanishing hash code gradient problem may occur during the hash encoding process. When the range of eigenvalues is too large or the distribution is uneven, the gradient may decrease gradually during backpropagation, making it difficult for the

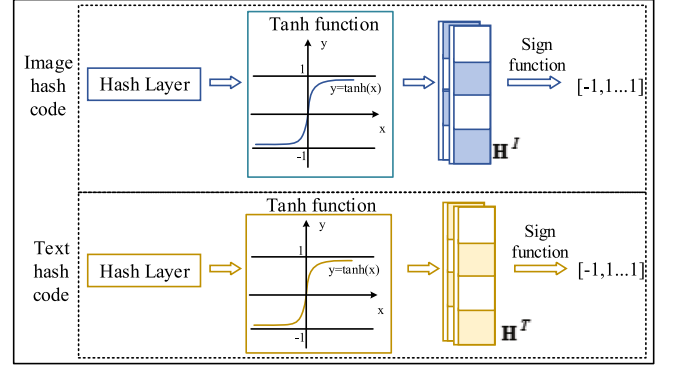


Fig. 3. The generation of hash code.

model to converge. To alleviate these problems, we introduce a $\tanh(\cdot)$ function after the hash encoding layer. The tanh function is a hyperbolic tangent function with an output range of $[-1, 1]$, which can map the input feature values to this interval, helping to smooth feature representation and avoid numerical instability. Additionally, the output range of the tanh function is symmetrical and continuous, which enables the model to update parameters more effectively during the back-propagation process, thereby accelerating model convergence. The following is a definition for the hash function:

$$\mathbf{H}^I = \tanh(\mathbf{W}^I \cdot \mathbf{I} + b^I), \quad (15)$$

$$\mathbf{H}^T = \tanh(\mathbf{W}^T \cdot \mathbf{T} + b^T), \quad (16)$$

where $\mathbf{W}^T$ and $\mathbf{W}^I$ denote the weights used for converting text and image features in the hash space, while b^I and b^T represent the corresponding bias terms. The $\tanh(\cdot)$ function still yields a continuous binary hash code. Through the usage of the $\text{sign}(\cdot)$ function, we can convert a continuous hash code to a discrete binary hash code. The definition for the $\text{sign}(\cdot)$ function is $\text{sign}(x)$, which is expressed as follows:

$$\text{sign}(x) = \begin{cases} 1 & \text{if } x \geq 0, \\ -1 & \text{if } x < 0, \end{cases} \quad (17)$$

where x represents the values of hash codes $\mathbf{H}^I$ and $\mathbf{H}^T$.

By using the $\text{sign}(\cdot)$ function, the eigenvalues can be discretized to obtain a discrete binary hash code. Discrete binary hash codes can directly calculate the Hamming distance between modalities, significantly improving retrieval efficiency and quickly finding similar features. The specific process of hash code generation is shown in Fig. 3.

2) *Loss Function*: We propose a technique that combines contrastive triplet loss and cosine similarity to improve hash code quality, yielding more semantically relevant hash codes and reduced quantization errors. Compared with the traditional contrastive loss, the triplet loss can more directly optimize the relative relationship between samples by introducing anchor points. By considering both positive and negative sample pairs, we further integrate the concept of contrastive loss, thereby improving the model's ability to distinguish between the similarities and differences of sample pairs. The following is a

description of cosine similarity:

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}, \quad (18)$$

where the hash code vectors for the image and text are indicated by $\mathbf{a}$ and $\mathbf{b}$, respectively. $\mathbf{a} \cdot \mathbf{b}$ denotes the dot product of vectors $\mathbf{a}$ and $\mathbf{b}$, $\|\mathbf{a}\|$ and $\|\mathbf{b}\|$ indicate their Euclidean norms (i.e., their lengths).

It is important to maximize the distance between the negative sample and the anchor sample and decrease the distance between the positive sample and the anchor sample for a triplet (anchor sample $\mathbf{a}$, positive sample $\mathbf{p}$, negative sample $\mathbf{n}$). The following is the particular contrastive triplet loss function:

$$\begin{aligned} \mathcal{L}_{\text{loss}} = & \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \sum_{k=1}^N (\mathbf{W}_{ijk} \cdot \text{mask}_{ijk} \cdot \\ & \max(0, \|f(\mathbf{a}_i) - f(\mathbf{p}_j)\|_2^2 - \\ & \|f(\mathbf{a}_i) - f(\mathbf{n}_k)\|_2^2 + \beta)), \end{aligned} \quad (19)$$

where $\mathbf{W}_{ijk}$ is the weight of a triplet, mask_{ijk} is a triplet mask used to filter out invalid triples, β is the margin parameter that is utilized to set the minimum distance gap between the positive and negative samples.

By combining contrastive triplet loss and cosine similarity, our method can better learn the similarities and dissimilarities between samples to enhance the ability of cross-modal retrieval. Furthermore, in our framework, the label classifier and the hashing module exhibit some robustness to label noise. Noise does not directly impact the hashing module. Labels are used to construct a co-occurrence graph based on conditional probabilities, which filters out weak or spurious connections. GCN and GAT further minimise unreliable relationships through neighborhood aggregation and adaptive attention. The resulting graph representations provide semantic guidance and are combined with modality features before hash learning, ensuring that the final hash codes are primarily shaped by multimodal feature similarity. This approach reduces sensitivity to noisy labels and enhances stability.

F. Optimization

In this section, we adopt an alternating optimization strategy to optimize the hash code. The alternating optimization strategy can more effectively capture the semantic relevance between different modal data, significantly reduce quantization errors, and improve the retrieval accuracy by gradually optimizing the generation process of hash codes. Through the alternating optimization strategy, the model can better learn the feature representation of different modal data and ensure that the generated hash codes are semantically consistent. Next, we will specifically introduce how we use the optimization strategy to optimize the hash code.

According to the EGATH [51] method for iteratively optimizing hash codes, the overall loss of our suggested approach is as follows:

$$\min_{\mathbf{R}, \mathbf{Z}, \mathbf{C}, \mathbf{H}} = \mathcal{L}_{\text{class}} + \mathcal{L}_{\text{loss}} + \omega \mathcal{L}_{\text{quant}} + \tau \mathcal{L}_{\text{reg}}, \quad (20)$$

$\mathcal{L}_{\text{quant}}$ is a quantization loss, e.g., $\|\mathbf{H} - \mathbf{B}\|_F^2$, $\mathbf{H}$ denotes the continuous hash representations, and $\mathbf{B}$ is the binary hash code matrix, ensuring minimal discretization error when mapping features to hash codes. $\mathcal{L}_{\text{reg}}$ is a regularization term on the model parameters. ω and τ are hyperparameters that control the relative weight of the quantization and regularization terms. High-dimensional feature vectors are mapped into a low-dimensional hash code space via a mapping matrix from the feature space to the hash code space, denoted by $\mathbf{R}$. The target hash code's ideal state, as close to the constraint conditions as feasible, is represented by $\mathbf{C}$. The difference between the target hash code $\mathbf{C}$ and the current hash code $\mathbf{H}$ is corrected using $\mathbf{Z}$. Through this alternating optimization strategy, the model can find the optimal hash code while satisfying the constraints. The entire optimization process is an iterative process that approximates the optimal hash code by gradually updating $\mathbf{R}$, $\mathbf{Z}$, $\mathbf{C}$, and $\mathbf{H}$.

Firstly, based on the current hash code $\mathbf{H}$, optimize the mapping relationship from features to hash codes, that is, update $\mathbf{R}$. The process is as follows:

$$\mathbf{R} = (\mathbf{L}^\top \mathbf{L} + \lambda \mathbf{E})^{-1} \mathbf{L}^\top \mathbf{H}. \quad (21)$$

Due to the update of $\mathbf{R}$, there is a deviation between the feature representation and the label data. For $\mathbf{H}$ to be nearer the intended hash code $\mathbf{C}$, we will update $\mathbf{H}$ to correct the deviation:

$$\mathbf{H} = \text{sign} \left(\mathbf{L}\mathbf{R} + \psi \left(\mathbf{C} - \frac{\mathbf{Z}}{\psi} \right) + \alpha(\mathbf{H}^I + \mathbf{H}^T) \right). \quad (22)$$

Since $\mathbf{C}$ is an ideal state hash code that satisfies the constraints that $\mathbf{H}$ should follow, the hash code $\mathbf{H}$ ought to be as close to the optimal state hash code as feasible. Using Singular Value Decomposition (SVD), we may break down the hash code $\mathbf{H}$ as follows:

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T. \quad (23)$$

The orthogonal matrices $\mathbf{U}$ and $\mathbf{V}$, which contain the left and right singular vectors, respectively, are in question. By further updating $\mathbf{C}$, the process is represented as follows:

$$\mathbf{C} = \mathbf{U}_C \mathbf{\Sigma}_C \mathbf{V}_C^T, \quad (24)$$

where $\mathbf{U}_C$ and $\mathbf{V}_C$ represent the columns extracted from matrices $\mathbf{U}$ and $\mathbf{V}$ that correspond to the largest singular values. Meanwhile, these chosen greatest singular values are contained in the diagonal matrix $\mathbf{V}_C^T$. By gradually correcting the deviation between $\mathbf{H}$ and $\mathbf{C}$ through $\mathbf{Z}$, when $\mathbf{H}$ equals $\mathbf{C}$, the update of $\mathbf{Z}$ is as follows:

$$\mathbf{Z} = \mathbf{Z} + \psi(\mathbf{H} - \mathbf{C}). \quad (25)$$

In each iteration, update $\mathbf{C}$ and $\mathbf{Z}$ alternately. First update $\mathbf{C}$ to reflect the constraint on hash code $\mathbf{H}$, and then adjust $\mathbf{Z}$ to correct the deviation from the target. When the change in hash code $\mathbf{H}$ falls below a predetermined threshold or exceeds the predetermined number of iterations, the loop should end.

Utilizing an alternating optimization technique can significantly improve hash code quality in cross-modal retrieval, thereby enhancing retrieval efficiency and accuracy. Through this alternating optimization strategy, find the best hash code. The overall algorithm is shown in Algorithm 1.

Algorithm 1: Learning Algorithm of our DGNH.

- 1: **Input:** Training set $\{\mathbf{x}_i, \mathbf{y}_i, \mathbf{I}_i \mid i \in [1, N]\}$, label set $\mathbf{L} = \{\mathbf{I}_i \mid i \in [1, N]\}$, hash code length $\mathbf{c}$, parameters α and β .
Output: hash codes $\mathbf{H}^I$ and $\mathbf{H}^T$.
- 2: **Initialization:** Initialize network parameters, hash codes $\mathbf{H}$, and hyper-parameters.
- 3: **repeat**
- 4: **for** each epoch **do**
- 5: Feature extraction operation to get $\mathbf{D}_I$:
 $f(x) \rightarrow \mathbf{D}_I$.
- 6: Feature extraction operation to get $\mathbf{D}_T$:
 $f(y) \rightarrow \mathbf{D}_T$.
- 7: Seed the label glove vector and adjacency matrix into the label classifier to get the output of feature representation $\mathbf{H}_i$ using (12): $(\mathbf{G}, \mathbf{A}) \rightarrow \mathbf{H}_i$.
- 8: Use the (13) to get the predicted score $\mathbf{Y}_i$ for the multi-label classification loss.
- 9: Obtain hash code through sign function and tanh function: $\mathbf{H}^I$ and $\mathbf{H}^T$.
- 10: Solve the overall loss function $\mathcal{L}$:
 $\mathcal{L} = \mathcal{L}_{class} + \mathcal{L}_{loss} + \omega \mathcal{L}_{quant} + \tau \mathcal{L}_{reg}$.
- 11: **end for**
- 12: Update $\mathbf{R}$ using (21).
- 13: Compute $\mathbf{H}$ using (22).
- 14: Use the (24) to calculate $\mathbf{C}$.
- 15: Use the (25) to calculate $\mathbf{Z}$ and ψ .
- 16: **until** convergence

TABLE II
DATASETS STATISTICS

Dataset	Size	Label	Query	raining	Text
MIRFlickr-25K	25,000	24	2,000	10,000	1386D
NUS-WIDE	269,648	81	2,100	10,500	1000D
MS-COCO	123,289	80	5,000	10,000	2026D

IV. EXPERIMENTS

A. Datasets

For our studies, we selected three well-known cross-modal datasets, the statistics of which are presented in Table II.

The MIRFlickr-25 K dataset [52] involves the 25,000 images (each tagged with at least one of 24 classes), and we retained only those having 20 or more tags. Out of these, 2,000 image-text pairs were randomly designated as the query set, the rest formed the retrieval set, and a further 10,000 pairs were randomly drawn from the retrieval set to create the training set.

The NUS-WIDE dataset [53] contains the 269,648 web images (each labeled with at least one of 81 classes and accompanied by text tags), and we selected 186,577 pairs belonging to the ten most frequent categories. From this subset, 2,100 pairs were randomly assigned to the query set, the remainder served as the retrieval set, and 10,500 pairs were randomly sampled from the retrieval set to form the training set.

The MS-COCO dataset [54] 123,289 images (each annotated with at least one of 80 classes). We randomly picked 5,000

image-text pairs to constitute the query set, used the remainder as the retrieval set, and then randomly sampled 10,000 pairs from that retrieval set for training.

B. Evaluation Metrics

Three main assessment metrics were employed in our experiments: the Precision-Recall curve (PR curve), the top-K precision curve (top-K curve), and mean Average Precision (mAP). The following is the definition of these metrics:

Mean Average Precision (mAP): The average precision (AP) of each query is computed, and the AP of all queries is then averaged to determine the mAP. It displays the model's total retrieval accuracy for various queries. The retrieval mechanism performs better when the mAP value is larger.

Precision-Recall Curve (PR Curve): In addition to naturally presenting the model's performance under high recall, it illustrates the relationship between the model's precision and recall under various thresholds. This is crucial for comprehending the model's accuracy in retrieving pertinent data.

Top-K Precision Curve (Top-K Curve): It evaluates the accuracy under various retrieval numbers, assesses the change in precision in the top K results based on the retrieval results, and helps in understanding the model's suitability in different contexts.

By combining these three indicators, the model's retrieval performance can be comprehensively evaluated and optimized to achieve better retrieval results.

C. Implementation Details

Our DGNH is powered by the NVIDIA GeForce RTX 3080 GPU with 20 GB of RAM and is implemented using PyTorch [55]. In our experiments, we employ CLIP and transformer as image feature extractors while extracting features from text using a fully connected layer and the transformer. We utilize GCN and GAT as the label classifier to model the dependencies between labels, capture the structural information in the graph, and generate more nuanced label representations. The input for the label classifier includes 300-D Glove vectors, as well as an adjacency matrix representing the relationships between labels.

We set some hyperparameters, denoted as α , β , λ , and μ , for auxiliary learning purposes. We have assessed the sensitivity of these hyperparameters through an experimental study. We selected the hyperparameters that yielded the best retrieval outcomes based on the experimental results. We obtain the expected label based on the classifier module and forward it to the hash module. To obtain the best hash code, we optimize the hash code in the hash module using the contrast triple loss function and alternating optimization approach. To optimize the entire network, we use Adam [56] with a learning rate of 10^{-4} . Lastly, we use a dynamic learning rate modification approach and set the batch size to 64.

D. Comparison With Baselines

We experimented with two retrieval tasks—"I $\rightarrow$ T" representing image-to-text retrieval and "T $\rightarrow$ I" representing text

TABLE III
THE MAP COMPARISON RESULTS ON THREE DATASETS

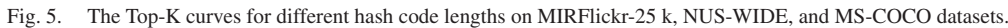
Task	Method	MIRFlickr-25K			NUS-WIDE			MS-COCO		
		16bits	32bits	64bits	16bits	32bits	64bits	16bits	32bits	64bits
I→T	DJSRH	0.6652	0.6873	0.6987	0.5271	0.5582	0.6015	0.5257	0.5454	0.5646
	JDSH	0.7276	0.7426	0.7468	0.6536	0.6601	0.6900	0.5928	0.6348	0.6517
	CDTH	0.7317	0.7461	0.7477	0.6596	0.6613	0.6700	0.5853	0.6411	0.6573
	UCCH	0.7606	0.7620	0.7674	0.6718	0.6738	0.6891	0.6039	0.6249	0.6398
	DSPH	0.7935	0.8141	0.8363	0.6851	0.6996	0.7092	0.6466	0.6666	0.6757
	MLCAH	0.7960	0.8080	0.8150	0.6440	0.6410	0.6430	0.5700	0.5620	0.5620
	SCAHN	0.8123	0.8131	0.8336	0.6588	0.6621	0.6669	0.6727	0.7108	0.7528
	DCHMT	0.8177	0.8221	0.8261	0.6711	0.6812	0.6932	0.6450	0.6331	0.6647
	DNPH	0.8101	0.8114	0.8100	0.6633	0.6790	0.6891	0.6094	0.6866	0.7016
	MITH	0.8192	0.8300	0.8329	0.6992	0.7135	0.7291	0.6959	0.7188	0.7582
	DFNSH	0.8081	0.8116	0.8126	0.6348	0.6577	0.6988	0.6390	0.6357	0.6059
	DNpH	0.8357	0.8513	0.8488	0.6854	0.6935	0.7004	0.6711	0.6683	0.6752
	EGATH	0.8411	0.8562	0.8637	0.7191	0.7345	0.7480	0.7259	0.7688	0.7945
	DGNH	0.8463	0.8573	0.8669	0.7193	0.7367	0.7482	0.7323	0.7753	0.8058
T→I	DJSRH	0.6710	0.6958	0.7043	0.5575	0.5680	0.5952	0.5590	0.5591	0.5519
	JDSH	0.7304	0.7326	0.7481	0.6439	0.6640	0.6921	0.5888	0.6510	0.6635
	CDTH	0.7315	0.7464	0.7503	0.6788	0.6815	0.6910	0.5846	0.6427	0.6573
	UCCH	0.7343	0.7342	0.7410	0.6740	0.6812	0.6945	0.6023	0.6258	0.6371
	DSPH	0.7928	0.8038	0.8149	0.6957	0.7147	0.7256	0.6473	0.6656	0.6774
	MLCAH	0.7940	0.8050	0.8050	0.6620	0.6730	0.6870	0.5440	0.5470	0.5940
	SCAHN	0.7890	0.7971	0.8186	0.6718	0.6803	0.6980	0.7183	0.7504	0.8093
	DCHMT	0.8007	0.8021	0.8065	0.6852	0.6963	0.7009	0.6298	0.6176	0.6616
	DNPH	0.7968	0.8171	0.8158	0.6811	0.6936	0.7098	0.6164	0.7012	0.7200
	MITH	0.8004	0.8163	0.8221	0.7135	0.7270	0.7306	0.6858	0.7089	0.7560
	DFNSH	0.7983	0.8003	0.8071	0.6691	0.6399	0.6415	0.6341	0.6285	0.5891
	DNpH	0.7977	0.8160	0.8251	0.7081	0.7054	0.7008	0.6380	0.6824	0.6830
	EGATH	0.8064	0.8185	0.8293	0.7270	0.7437	0.7539	0.7247	0.7729	0.8015
	DGNH	0.8069	0.8224	0.8314	0.7298	0.7492	0.7592	0.7340	0.7780	0.8116

-to-image retrieval—and compared the outcomes on three datasets: MIRFlickr-25 K, NUS-WIDE, and MS-COCO to evaluate the efficacy of the proposed DGNH model. The baseline models consist of 13 advanced cross-modal hashing methods that include Deep Joint-semantics Reconstructing Hashing (DJSRH) [57], Joint-modal Distribution-based Similarity Hashing (JDSH) [58], Cross Domain Transfer Hashing (CDTH) [10], Unsupervised Contrastive Cross-modal Hashing (UCCH) [40], Deep Semantic-aware Proxy Hashing (DSPH) [59], Multi-level Correlation Adversarial Hashing (MLCAH) [60], Structures Aware Fine-grained Contrastive Adversarial Hashing (SCAHN) [61], Differentiable Cross-modal Hashing via Multimodal Transformers (DCHMT) [62], Deep Neighborhood-aware Proxy Hashing (DNPH) [63], Multi-granularity Interactive Transformer Hashing (MITH) [64], Deep Feature-based Neighbor Similarity Hashing (DFNSH) [65], Deep Neighborhood-preserving Hashing (DNpH) [66] and An End-to-End Graph Attention Network Hashing (EGATH) [51]. To maintain fairness, we re-executed the code for the comparative experiments. For these papers that did not provide their code, we directly utilized the results published in the papers, ensuring that the dataset splits were consistent with those reported. To evaluate the retrieval performance, we choose top-K precision curves, PR curves, and mAP as assessment criteria.

As shown in Table III, the mAP results for each method are reported for hash code lengths of 16, 32, and 64, with the best values highlighted in bold. Several observations can be made. Comparative experiments on three benchmark datasets indicate that the proposed method consistently outperforms other competing approaches in overall performance. It achieves stable advantages

on both small- and medium-scale datasets, demonstrating strong generalization and robustness. Notably, the improvement is most pronounced on the MS-COCO dataset, where the diversity, complex semantic relationships, and large-scale pose present greater challenges. This confirms the effectiveness of the method in complex cross-modal retrieval scenarios. Specifically, compared to the baseline method with the lowest mAP result, our method improved performance by 20.9%, 22.9%, and 24.1% for three-bit hash codes in image-to-text retrieval, and by 17.5%, 21.8%, and 25.9% for three-bit hash codes in text-to-image retrieval. This substantial improvement can be attributed to the richer label information in MS-COCO, which provides more contextual cues and enhances both label prediction accuracy and cross-modal feature consistency. However, for the NUS-WIDE dataset, the performance gain of our method over the top-performing baseline is relatively modest. This is because the NUS-WIDE dataset, with its sparse labels, limited textual semantic information, and uneven class distribution, fails to fully realize the potential of our method in cross-modal feature alignment, thus resulting in a relatively limited performance improvement.

Fig. 4 presents the PR curves for the three datasets using hash codes of 16, 32, and 64 bits in length. The PR curves of our proposed approach outperform those of alternative approaches in the majority of instances. This demonstrates that our method can maintain high accuracy across various recall levels, particularly in high recall areas, where the model's accuracy is more pronounced. For complex datasets, our method has an advantage over other methods. The feature network in DGNH effectively amplifies the semantic distinction across modalities



The findings of all three evaluation criteria for the top K, PR, and mAP precision curves indicate that our approach has outperformed the state-of-the-art techniques to some extent. Our method utilizes CLIP and Transformer-based encoders, in contrast to earlier cross-modal hashing methods (e.g., DJSRH, JDSH, and CDTH), which mainly rely on CNN-based visual

features and shallow text representations. CLIP is trained on large-scale image-text pairs and provides naturally aligned multimodal representations, while the Transformer module further refines the embeddings and captures global semantic dependencies. This leads to greater semantic consistency across modalities, thereby directly benefiting cross-modal retrieval. Second, many existing methods focus primarily on instance-level similarity learning, while DGNH additionally models label-level semantic relationships. By constructing a label association graph and applying the GCN-GAT classifier, the model can capture

TABLE IV
ABLATION STUDY ON THE THREE DATASETS

Task	Method	MIRFlickr-25K			NUS-WIDE			MS-COCO		
		16bits	32bits	64bits	16bits	32bits	64bits	16bits	32bits	64bits
I→T	DGNH-GCN	0.7326	0.7284	0.7803	0.6693	0.6936	0.7075	0.6085	0.6213	0.6287
	DGNH-GAT	0.6229	0.7358	0.7345	0.6690	0.6862	0.6945	0.6073	0.6292	0.6364
	DGNH-SG	0.8369	0.8546	0.8612	0.7064	0.7189	0.7425	0.7195	0.7574	0.7821
	DGNH-CT	0.8406	0.8543	0.8616	0.7125	0.7310	0.7433	0.7131	0.7559	0.7882
	DGNH	0.8463	0.8573	0.8669	0.7193	0.7367	0.7482	0.7323	0.7753	0.8058
T→I	DGNH-GCN	0.7223	0.7253	0.7656	0.6855	0.7072	0.7163	0.6084	0.6191	0.6270
	DGNH-GAT	0.5592	0.7290	0.7363	0.6774	0.7063	0.7052	0.6030	0.6263	0.6394
	DGNH-SG	0.8014	0.8133	0.8238	0.7189	0.7304	0.7523	0.7249	0.7623	0.7892
	DGNH-CT	0.7966	0.8162	0.8237	0.7224	0.7372	0.7483	0.7144	0.7610	0.7932
	DGNH	0.8069	0.8224	0.8314	0.7298	0.7492	0.7592	0.7340	0.7780	0.8116

correlations and dependencies among labels. This graph-based propagation enables the model to incorporate structured semantic information beyond individual samples, thereby improving the quality of semantic supervision during hash learning. Finally, the contrastive triplet loss combined with quantization-aware optimization encourages semantically similar cross-modal samples to obtain closer hash codes while reducing the quantization gap between continuous representations and binary codes. This leads to more stable and discriminative hash codes for large-scale retrieval. These factors jointly contribute to the performance improvements observed in the experiments. In particular, the advantage becomes more evident on datasets with rich semantic annotations and complex label relationships, where modeling label correlations and contextual semantics plays a more significant role.

E. Ablation Study

The outstanding performance of the DGNH approach in cross-modal hashing retrieval was validated by comparing trials with several cross-modal hashing techniques in earlier studies. We also experimented with several variations of the DGNH method on three datasets: MIRFlickr-25 K, NUS-WIDE, and MS-COCO, to confirm the method's contribution. We conducted two ablation experiments as follows:

- *DGNH-GCN*: In this experiment, we removed the Graph Convolutional Network module while keeping all other components identical to those in DGNH.
- *DGNH-GAT*: In this experiment, we removed the Graph Attention Network module while keeping all other components identical to those in DGNH.
- *DGNH-SG*: In this experiment, we removed the GCN and GAT classifier module while keeping all other components identical to those in DGNH.
- *DGNH-CT*: In this experiment, we removed the contrastive triplet loss function based on cosine similarity while maintaining all other components the same as in DGNH.

Table IV presents the ablation study results, highlighting the contribution of each key module to overall performance. Removing any module reduces retrieval accuracy, confirming their complementary roles. Removing GCN or GAT leads to a significant drop—for 16-bit image-to-text retrieval on MIRFlickr-25 K, the full DGNH achieves 0.8463, whereas DGNH-GCN and DGNH-GAT drop to 0.7326 and 0.6229, respectively. GCN

preserves global structural consistency, while GAT enhances adaptive feature aggregation, and their integration enables more effective cross-modal representation learning. The introduction of the classifier module based on the GCN and GAT significantly enhances the structural modeling and semantic capture capabilities of cross-modal features, further bridging the semantic gap. Furthermore, the contrastive loss optimization strategy employed in the hash code generation module effectively improves feature alignment and discrimination between different modalities, significantly enhancing cross-modal retrieval performance. Experiments on three benchmark datasets demonstrate that integrating these modules significantly outperforms control methods in retrieval tasks, with the most significant improvement achieved on the MS-COCO dataset. This not only validates the effectiveness of the proposed method in scenarios with large-scale and diverse cross-modal data but also demonstrates the synergistic effects and advantages of each module in complex semantic matching tasks. Overall, the classifier module, which combines the GCN and GAT, and the optimized contrastive loss in the hash code module, not only enhances feature extraction and semantic alignment but also delivers notable performance gains in cross-modal retrieval tasks.

F. Parameter Sensitivity

In this study, we optimized several key parameters and conducted parameter sensitivity analysis by systematically adjusting their values and observing the impact on cross-modal retrieval performance. The experimental results show that mAP fluctuates with parameter changes, reflecting the influence of parameter selection on model performance and confirming the rationality of the chosen settings. This analysis further helps reveal the contribution of each parameter and facilitates optimal parameter selection. Specifically, set α to 1 and β to 0.3.

We established several parameters for experimental optimization in this investigation. To evaluate these parameters, we performed validation across three datasets using a consistent 64-bit hash code. When evaluating one parameter, the other parameters are fixed. Fig. 6 shows the specific results. As we adjusted the values of these parameters, the mAP results exhibited fluctuations, which confirmed the rationality of our parameter choices for the model. The optimal mAP values were achieved in the three data sets when α was set to 1. Regarding β , as its value increased from 0.1 to 0.3, the performance

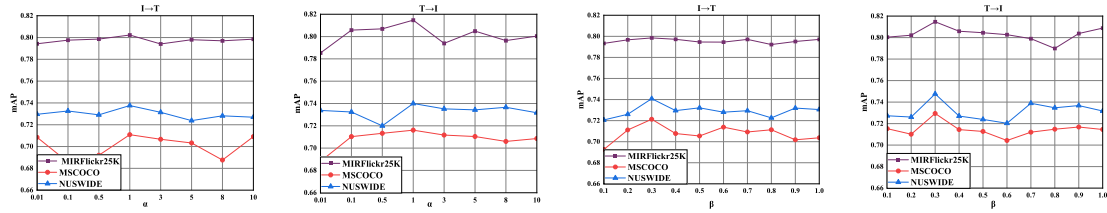


Fig. 6. The influence of hyper-parameters.

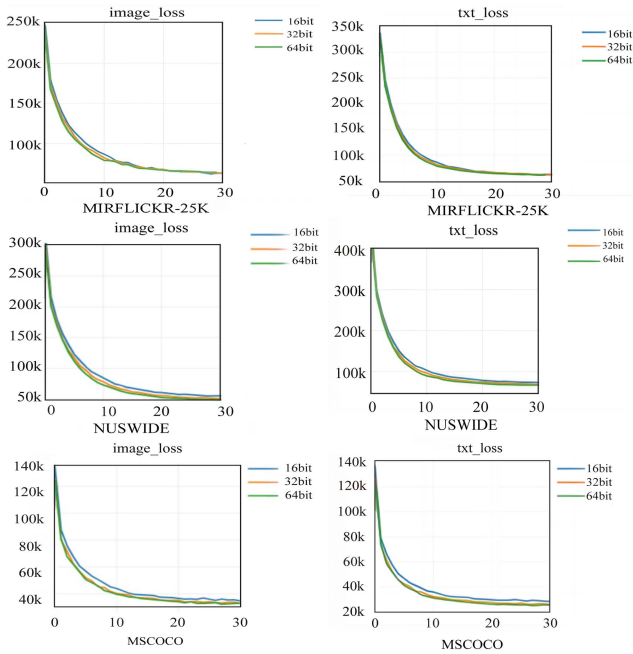


Fig. 7. Convergence curves of images and text on three datasets.

metrics improved gradually, indicating a positive impact on regularization. After increasing progressively from 0.3, the performance metrics decreased. Therefore, we set β to 0.3.

G. Convergence Analysis

To evaluate the stability and efficiency of the optimization process, we plotted the convergence curves of image and text losses on three benchmark datasets, as shown in Fig. 7. It can be observed that on all datasets, both the image and text losses decrease rapidly at the beginning of training and then gradually level off, demonstrating that the proposed method achieves efficient and stable convergence. Most loss reduction occurs within the first 5–10 epochs, after which the curves level off, indicating that the model has essentially converged. Moreover, the convergence behavior under different hash code lengths is highly consistent, with nearly overlapping curves and similar final loss values, demonstrating the robustness of the proposed method to hash code length variations. Although MIRFLICKR-25 K and NUSWIDE exhibit relatively higher initial losses, especially in the text modality, the model still converges smoothly. In contrast, MSCOCO shows lower initial

loss, faster convergence, and lower final loss, indicating better adaptation to datasets with richer annotations and more complex semantic structures.

H. Inference Efficiency and Computational Cost

In our method, the edges of the label association graph are constructed based on label co-occurrence statistics from the training set. The graph construction mainly involves computing the label co-occurrence matrix and conditional probabilities. Since the number of labels is typically much smaller than the number of training samples and the graph is built only once before training, the associated computational overhead is negligible compared to the overall training process. In addition, the main computational cost comes from CLIP and Transformer-based feature extraction, while the GCN-GAT module and alternating optimization introduce only limited additional overhead. Training was conducted on the NVIDIA GeForce RTX 3080 GPU with 64-bit hash codes and a batch size of 64. For the MIRFlickr-25 k dataset, each epoch required approximately 3 minutes, with peak GPU memory usage around 2.73 GB and a maximum GPU utilization of 80%. On the NUS-WIDE dataset, training time increased to about 7.2 minutes per epoch, with peak memory usage of approximately 2.64 GB and peak GPU utilization of 77%. For the MS-COCO dataset, each epoch took roughly 11.2 minutes, with a peak memory consumption of 3.47 GB and a maximum GPU utilization of 55%.

V. CONCLUSION

We suggest DGNH for cross-modal retrieval in this paper. First, DGNH extracts characteristics from text and images using the Transformer and CLIP model. It ensures semantic coherence between text and images, and enhances the overall feature representation of multimodal data through an end-to-end design. To help the model focus on the modality's semantic information and comprehend the semantic content more precisely, a contextual attention mechanism is also implemented simultaneously. Second, DGNH employs GCN and GAT to construct labels, enabling the model to uncover hidden features and relationships in the data and learn more about the semantic relationships between various modalities, making it possible to bridge the information gap that separates various data kinds. Additionally, DGNH employs a contrastive triplet loss function and an alternating optimization strategy to determine the optimal hash code, thereby enhancing the efficiency and performance of cross-modal retrieval. The method's good performance is

finally shown by experimental findings on three datasets. Although DGNH achieves strong retrieval performance, it still has certain limitations. First, due to the computational complexity of GCN and GAT, the model may incur additional overhead, potentially affecting retrieval efficiency. Moreover, the performance of DGNH relies on the quality of label information used for graph construction; noisy or inaccurate labels may weaken the learned semantic relationships and consequently degrade retrieval accuracy. In this work, we focus on supervised settings with reliable annotations. However, extending DGNH to handle noisy, incomplete, or weakly supervised scenarios is an important direction for future research. In addition, exploring more efficient architectures and adapting the framework to broader multimodal data, such as video and audio, could further enhance both scalability and practical applicability in real-world environments. This direction is also motivated by a limitation of the current study, as DGNH is evaluated only on image-text data and relies on CLIP encoders that do not capture temporal or sequential information. Nevertheless, owing to the modular design of the framework, it can be naturally extended by incorporating modality-specific encoders (e.g., for video or audio), while retaining the graph-based semantic modeling and hashing components.

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Shuang Zhang received the PhD degree in information and communication engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2020. She is currently an associate professor with the School of Engineering College, Hebei Normal University. She has authored more than ten peer-reviewed papers in IEEE journals and conferences. Her research interests include 6G wireless networks, RIS, and AI-driven communication optimization. She was also a reviewer for multiple premier AI conferences, including NeurIPS and AAAI in 2024 and 2025.



Yue Wu received the BS degree from the School of Electronic and Information Engineering, Nanjing University of Information Science and Technology, Jiangsu, China, in 2022. She is currently working toward the MS degree with the College of Engineering, Hebei Normal University. Her research interests include cross-modal retrieval, hashing learning, and machine learning.



Lei Shi received the PhD degree from the School of Computer Science, Beijing University of Posts and Telecommunications, Beijing, China, in 2019. He is currently an associate professor with the State Key Laboratory of Media Convergence and Communication, Communication University of China. His research interests include artificial intelligence, data mining, and cross-media search.



Feifei Kou received the PhD degree from the School of Computer Science, Beijing University of Posts and Telecommunications, in 2019. She is currently an associate professor with the School of Computer Science (National Pilot School of Software Engineering), Beijing University of Posts and Telecommunications, China. From 2019 to 2021, she was a postdoctoral researcher with the School of Computer Science (National Pilot School of Software Engineering), Beijing University of Posts and Telecommunications. Her research interests include semantic learning, and multimedia information processing.



Huilong Jin received the MS and PhD degree in circuits and systems from Yanshan University, in 2000 and 2012, respectively. He is currently a full-time professor with Hebei Normal University, China. His research interests include machine learning, digital image processing, and communication theory.



Mingying Xu received the PhD degree from the School of Computer Science, Beijing University of Posts and Telecommunications, in 2022. She is currently a lecturer with the School of Artificial Intelligence and Computer Science, North China University of Technology. Her research interests include artificial intelligence, natural language processing, and multimodal learning.



Pengfei Zhang (Associate Member IEEE) received the PhD degree majoring in computer science from the Beijing University of Posts and Telecommunications in China. He has authored or coauthored more than 40 papers in *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Mobile Computing*, *NeurIPS*, *ICML*, and *AAAI*. His research interest focuses on privacy protection in mobile crowdsensing. He was the reviewer, meta-reviewer or area chair for many top-tier conferences and journals.



Muhammet Deveci received the BSc degree in industrial engineering from Cukurova University, Adana, Turkey, in 2010, the MSc degree in business administration from Gazi University, Ankara, in 2012, and the PhD degree in industrial engineering from Yildiz Technical University, Istanbul, Turkey, in 2017. During 2014–2015 and 2018–2019, respectively, he was a visiting researcher and postdoctoral researcher with the School of Computer Science, University of Nottingham, Nottingham, U.K. He is currently an honorary senior research fellow with the Bartlett School of Sustainable Construction, University College London, U.K., and also a full professor with the Department of Industrial Engineering, Turkish Naval Academy, National Defence University, Istanbul, Turkey. He is also an outstanding researcher and a prolific author who has been publishing high quality peer-reviewed papers in highly ISI ranked journals and reputable international conferences. He has authored or coauthored more than 415 papers in journals indexed by SCI/SCI-E papers at reputable venues, and more than 30 contributions in International Conferences related to his areas. Dr. Deveci was the recipient of the 100th-Anniversary Award for his worldwide scientific achievements from the Scientific and Technological Research Council of Turkey (TUBITAK).



Weiping Ding received the PhD degree in computer science from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2013. In 2016, he was a visiting scholar with the National University of Singapore, Singapore. From 2017 to 2018, he was a visiting professor with the University of Technology Sydney, Australia. He is currently a full professor with Nantong University. He has authored or coauthored more than 480 articles, including more than 260 IEEE Transactions papers. His research interests mainly include granular data mining and multimodal machine learning. He is also an associate editor/area editor/editorial board member of more than ten international prestigious journals, such as *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Fuzzy Systems*, *IEEE Transactions on Intelligent Transportation Systems*, *Information Fusion*, *Neurocomputing*, and *Applied Soft Computing*.